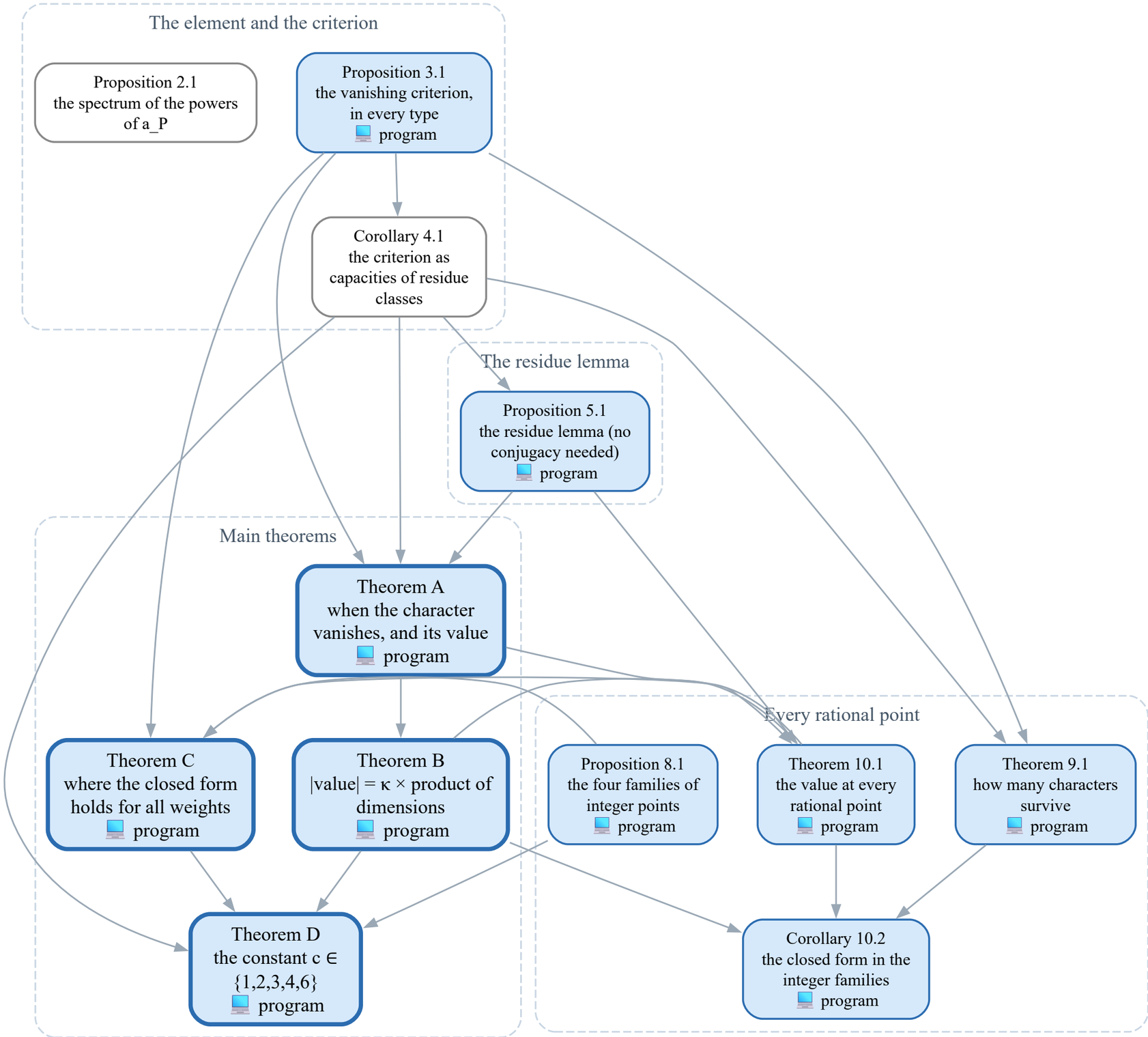
Map of *Characters of Sp(2m) at the powers of Kostant's element*

Representation theory · characters of Lie groups **05E10** 20G05 17B22 05A15

12 results | **0** formalized in Lean | ◐ **0** with a part in Lean | **10** checked by a program | written proof only (a result can carry several tags)

Dependencies in the written proof: an arrow A → B means that this proof of B uses A. Every result rests on its written proof; the tags record what was checked besides, and each statement's "Evidence & scope" block says how far. The colour of a box is its strongest tag, in the order above; the tags in the box give the rest. Boxes with a thick border are the main results; click a box to read the statement.


The element and the criterion
Proposition 2.1
the spectrum of the powers of a_P
Proposition 3.1
the vanishing criterion, in every type
program
Corollary 4.1
the criterion as capacities of residue classes
The residue lemma
Proposition 5.1
the residue lemma (no conjugacy needed)
program
Main theorems
Theorem A
when the character vanishes, and its value
program
Theorem C
where the closed form holds for all weights
program
Theorem B
|value| = κ × product of dimensions
program
Theorem D
the constant c ∈ {1,2,3,4,6}
program
Every rational point
Proposition 8.1
the four families of integer points
program
Theorem 10.1
the value at every rational point
program
Theorem 9.1
how many characters survive
program
Corollary 10.2
the closed form in the integer families
program

What the paper contributes

Kostant's element a_P is a point where every irreducible character of a compact simple group takes a value in $\{0, \pm 1\}$. The paper determines the irreducible characters of Sp(2m) at **every power** a_P^k.

- **When a character vanishes, and its value** (Theorem A): it vanishes exactly when $\lambda+\rho$ lies on more walls of the modular root arrangement than ρ does, equivalently when some folded residue class holds more entries than its capacity; otherwise the value is an explicit signed ratio of root products.
- **The absolute value is a product of dimensions** of representations of GL, Sp and odd SO, times $\kappa \in \{1, 2\}$ (Theorem B).
- **Every rational point of the same line:** a non-vanishing character is, up to sign and that factor, such a product times a character of Sp(2s) or SO(2s), $s \leq m$, at a regular element (Theorem 10.1); the points where the closed form holds for all weights are listed (Theorem C), and at the points where all characters are integers the remaining factor is 1, 2, 3, 4 or 6 (Theorem D).

Known and new, as the paper states it. For k prime to 2m+2, Theorem A is due to Cellini, Möseneder Frajria and Papi; for gcd(k, 2m+2) odd, Theorems A and B follow from Albion's factorisations of universal characters, with the sign added. New: for gcd(k, 2m+2) even and $q \geq 3$, the values on the classes not conjugate to that of ρ and the factor κ; the list of points in Theorem C and the exclusion of the others; in Theorem D, the constants for $q \geq 4$ even with $q \mid 2m$, $q \nmid m$, and for $q = 6$ with $m \equiv 4 \pmod 6$.

Key tool: an identity of residue multisets (Proposition 5.1), which takes the place of the conjugacy hypothesis in the work of Nadimpalli, Pattanayak and Prasad.

🚀 **Start here:** **Theorem A**: when the character vanishes, and its value.

Main results:

- **Theorem A**: when the character vanishes, and its value
- **Theorem B**: |value| = κ × product of dimensions
- **Theorem C**: where the closed form holds for all weights
- **Theorem D**: the constant $c \in \{1,2,3,4,6\}$

🧭 Fields: where the paper lives and what it connects

Tools taken from other branches

- **Root systems** → walls of the modular root arrangement: the count criterion, proved by the argument of Nadimpalli–Pattanayak–Prasad (the alcove argument of Cellini–Möseneder Frajria–Papi needs ρ regular and no longer applies) — Proposition 3.1, Theorem A
- **Elementary number theory** → residues modulo q in folded classes; an identity of residue multisets — Corollary 4.1, Proposition 5.1
- **Algebraic combinatorics** → partitions, Weyl dimension formulas; the factorisations of universal and classical characters (Albion, Ayyer–Kumari) — Theorem B
- **Algebraic number theory** → Galois action on the spectrum: which points give integer characters (a lemma of Gannon–Schopieray) — Proposition 8.1
- **Enumeration** → counting the non-vanishing residues — Theorem 9.1

What the results serve

- **Characters at elements of finite order** ← Kostant's $\{0, \pm 1\}$ theorem, extended to every power of a_P; the question of Nadimpalli, Pattanayak and Prasad on vanishing and values at torsion elements, addressed for these elements of Sp(2m) — Theorem A, Theorem B
- **Reeder's constants** ← their values on the classes not conjugate to that of ρ — Theorem A, Proposition 5.1
- **Factorisations of classical characters** ← a factor κ and a residual character of type C or D at every rational point of the line — Theorem B, Theorem 10.1, Theorem D